Existential risk from advanced AI: Comment on the NTIA AI accountability policy

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Jun 9, 2023

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Commenters are "invited to provide input on relevant questions not asked", and so, before addressing other questions, we would like to address this question:

A. Are there any kinds of advanced AI that would plausibly kill everyone if built, no matter the intentions of the designers?

Yes. Before we discuss the details of which kinds, we'll assure the reader that this is a mainstream position.

Four recent Turing award recipients (the computer science equivalent of the Nobel) have expressed concern—Geoff Hinton, Yoshua Bengio, Judea Pearl, and Joseph Sifakis. Hinton expects that certain kinds of AI would seek control, because "Getting more control is a very good subgoal, because it helps you achieve other goals" (Hinton, 2023). Asked about the result,

[1] Our qualifications to write this are as follows. Michael Cohen has published technical research on the topic of advanced AI at several top AI journals and conferences, and this was the topic of his thesis. The publication venues include the Journal of Machine Learning Research, the Conference on Learning Theory, the conference of the Association for the Advancement of Artificial Intelligence, AI Magazine, and the Journal of Selected Areas in Information Theory. Michael Osborne is professor of machine learning at Oxford and has commented extensively on the topic of advanced AI in various public settings. Both have been invited to Westminster to testify to the House of Commons Science and Technology Select Committee on the topic of existential risk from advanced AI.

he said “I think it is quite conceivable that humanity is just a passing phase in the evolution of intelligence… It may keep us around for a while to keep the power stations running, but after that, maybe not.” From Bengio, “rogue” AI could “potentially endanger our societies and even our species” (Bengio, 2023). Pearl has endorsed the view represented by the book of another prominent computer scientist, Stuart Russell, which discusses the serious challenge of creating advanced AI which is (euphemistically) “compatible” with human life (Russell, 2019). Sifakis, along with Bengio, along with tens of thousands of non-Turing award winners, signed the Future of Life Institute’s open letter recommending a pause to the development of advanced AI systems (Future of Life Institute, 2023). Alan Turing himself considered the prospect of very intelligent computers, and concluded, “At some stage we should have to expect the machines to take control” (Turing, 1951).

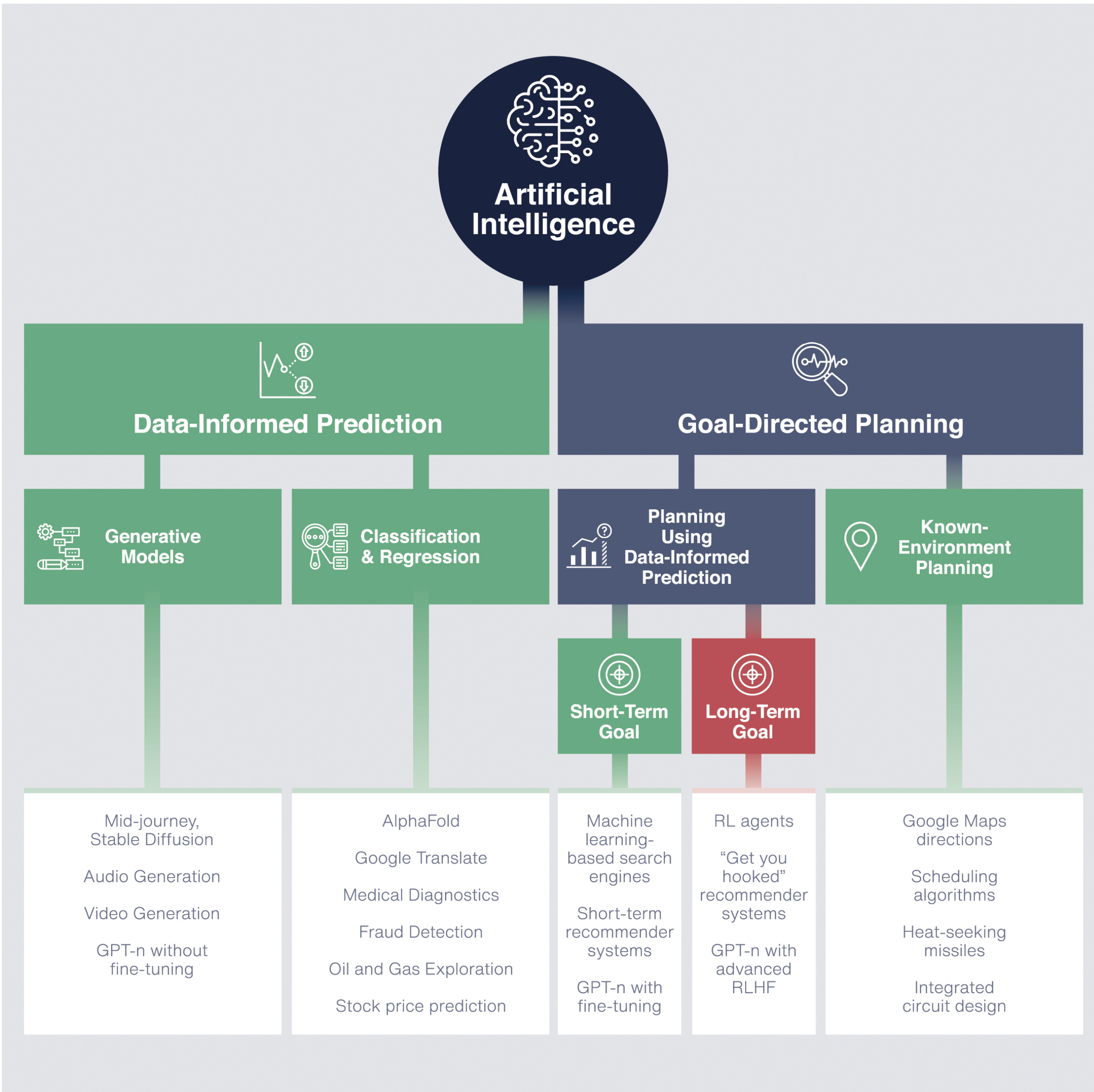
Sam Altman, CEO of OpenAI, has said, “AI will probably most likely lead to the end of the world” and that “development of superhuman intelligence is probably the greatest threat to the continued existence of humanity”. If there is any doubt about his meaning, he later says that superhuman machine intelligence “does not have to be the inherently evil sci-fi version to kill us all. A more probable scenario is that it simply doesn’t care about us much either way, but in an effort to accomplish some other goal (most goals, if you think about them long enough, could make use of resources currently being used by humans) wipes us out” (Altman, 2015a; Altman, 2015b). Dario Amodei, CEO of Anthropic, was lead author on a paper (Amodei, 2016) that contemplates the “risk of misspecified objective functions in superintelligent agents [Bostrom, 2014]”; this citation is to a book whose thesis is that such a scenario could be fatal to humanity. Shane Legg, co-founder of DeepMind, has said that the chance of “human extinction … within a year of something like human-level AI" was "maybe 5%, maybe 50%" (depending in part on what his interlocutor meant by human-level AI) (Legg, 2011).[2] A recent statement signed by the executives above, alongside Demis Hassabis, CEO of Google DeepMind, Ilya Sutskever, co-founder and chief scientist at OpenAI, Eric Horvitz, chief scientific officer at Microsoft, Kevin Scott, CTO at Microsoft, and many professors of computer science, says: "Mitigating the risk of extinction from A.I. should be a global priority alongside other societal-scale risks, such as pandemics and nuclear war" (Center for AI Safety, 2023).

We'll now briefly review the academic literature into which kinds of AI present this threat, why, and how. Cohen et al. (2022) find that under certain conditions, a *sufficiently advanced* artificial agent, one that is much more strategic than a human, would likely aim to intervene in its feedback and pursue arbitrary control over the world's resources in order to protect its ability to continue to control its own feedback. This would likely be fatal for humanity, again under those conditions, given the agent's appropriation of resources that we need to survive. Recent research (Turner et al., 2021) has formally verified the intuitive claim that for a random goal, seeking power is likely to be a component of optimal behavior. Zhuang and Hadfield-Menell (2020) model a resource-constrained world, and they find that if any expensive values do not feature in an agent's goal, they will likely be trampled once the agent becomes sufficiently advanced. This would certainly apply to an agent merely maximizing the probability of seeing certain observations, which is a common AI design. See also Russell (2019), Pan et al. (2022), and Carlsmith (2022).

[2] Lest the reader think these entrepreneurs cannot possibly believe what they are saying, or else they would not be advancing AI themselves, many express that they believe that a race to advanced AI is inevitable and that our best bet of surviving is with them at the helm.

Those sources discuss why an advanced artificial agent might pursue control over resources in a way that causes human extinction, but how would it do so? On this, we can only speculate. Perhaps it could manipulate people into giving it access to an internet terminal if it does not already have it. Perhaps it could develop novel pathogens or hack into weapons systems. However, this sounds a bit like chimpanzees saying, "Humans seem to be more intelligent, so maybe they could wipe us out by making bigger sticks to wield or stealing ours when we're not looking." What we have actually done to threaten chimpanzee's habitat would have been inconceivable to them. There is little reason to doubt that a sufficiently advanced AI would be able to design and control self-replicating nanotechnology and shape the world how it likes with atomic precision (Drexler, 1992 and personal conversation). However speculative these individual possibilities are, it would be foolish to bet on a very intelligent agent's inability to overcome human control.

Luckily, the cited research only concerns one kind of AI: artificial agents using data-informed prediction to plan actions over the long-term. The relevant taxonomy of AI, for determining the level of extinction risk, is depicted below (reproduced from a manuscript under review). Note that the taxonomy is relevant for evaluating extinction risk from AI that is much more advanced than what exists today, but we use current applications to illustrate the differences between categories of AI. In the interest of brevity, we will not discuss all the cells in the diagram—most are quite intuitive, and the examples may help. But we'll clarify the difference between "known environment planning" and "planning using data-informed prediction". To plan actions toward a goal, an agent must understand the consequences of its actions. Sometimes, these consequences are simple enough that designers can provide the agent with simple code that computes the consequences of its actions; this situation is called a known environment. But often, the world is too complex for designers to do this, so the agent must learn to model the consequences of its actions using data.



That fact that not all AI presents a high extinction risk makes it much easier to craft targeted regulation that allows a broad ecosystem of economically valuable AI to flourish.

Turning now to some questions asked by the NTIA,

1. What is the purpose of AI accountability mechanisms such as certifications, audits, and assessments?
    a. What kinds of topics should AI accountability mechanisms cover? How should they be scoped?

AI accountability mechanisms should have many purposes, addressing the many risks from AI. One of the should be the prevention of human extinction.

    e. Can AI accountability practices have meaningful impact in the absence of legal standards and enforceable risk thresholds? What is the role for courts, legislatures, and rulemaking bodies?

Probably not. Legislatures must write a framework for rulemaking bodies to develop rules to discern and ban the most dangerous forms of AI—very advanced agents planning over the long term. Rulemaking bodies would have to develop clear proxies for AI capability, such as the amount of compute used to train the AI, and also clarify a procedure for recognizing whether an algorithm produces a long-term planning agent, if the statutory text leaves any doubt. If criminal penalties are imposed, there may be a role for juries, informed by technical experts, to judge whether an algorithm produces a long-term planning agent.

2. Is the value of certifications, audits, and assessments mostly to promote trust for external stakeholders or is it to change internal processes? How might the answer influence policy design? How might the answer influence policy design?

To change internal processes! This makes robust enforceability a key concern.

3. AI accountability measures have been proposed in connection with many different goals, including those listed below. To what extent are there tradeoffs among these goals?
    d. The AI system is legal, safe, and effective.

If “safe” is understood to include “not causing human extinction”, there could indeed be tradeoffs with other objectives. Reinforcement Learning from Human Feedback (RLHF) is technique used by OpenAI to discourage ChatGPT from behaving in undesired ways, including hallucination of made-up facts. However, this method imbues an otherwise purely predictive model with long-term goals (such as avoiding human disapproval). If ChatGPT were more advanced, modifications that introduce long-term goals would introduce a risk of human extinction—for instance, ChatGPT might realise it will never be met with disapproval if all humans are dead.

It would be best to specifically name human extinction as a risk, rather than having it as an implicit component of “unsafe behavior,” because risk mitigation may take a different form than for other kinds of safety, and it should be more likely to supersede conflicting concerns.

11. What lessons can be learned from accountability processes and policies in cybersecurity, privacy, finance, or other areas?

Have there been any accountability processes in other domains that are so precisely constructed and so carefully enforced that no consequential violation has ever occurred? If so, these are the processes we need to learn from.

15. The AI value or supply chain is complex, often involving open source and proprietary products and downstream applications that are quite different from what AI system developers may initially have contemplated. Moreover, training data for AI systems may be acquired from multiple sources, including from the customer using the technology. Problems in AI systems may arise downstream at the deployment or customization stage or upstream during model development and data training.
    a. Where in the value chain should accountability efforts focus?

The government needs to know if anyone tries to build a sufficiently-capable long-term planning agent. Therefore, accountability processes must occur (at least) at the stage of AI system development.

Accountability may also be helpful further upstream, especially if such development becomes cheap enough for small organization or individuals to manage. The US could place reporting requirements on cloud computing providers, or even require chip manufacturers to add monitoring technology at a hardware level. Shavit (2023) conducts a preliminary investigation into how regulators could use such methods to supervise AI development.

Alternatively, it may suffice for cloud computing providers to share with auditors the bank account information and expenditures of their clients. And it may suffice for high-performance chips to require registration, whoever their owner, as with dangerous weapons. The government may decide that a security clearance is an appropriate requirement for owning high-performance chips.

Such upstream measures could also allow the US government to protect US citizens from AI developers abroad, if they use computing infrastructure that is managed or built by companies subject to US law.

d. Since the effects and performance of an AI system will depend on the context in which it is deployed, how can accountability measures accommodate unknowns about ultimate downstream implementation?

For sufficiently-advanced long-term planning agents, the risk of extinction is unacceptably large no matter the context in which it is deployed. So these unknowns have no bearing on regulation. On the other hand, usage of an advanced predictive algorithm as a subroutine of an artificial agent would introduce extinction risk. Such a user should be treated as an AI developer. Thus, AI developers that allow others to use their predictive models could assist auditors in identifying other AI developers who may need to be investigated, just as cloud computing providers could do the same.

16. The lifecycle of any given AI system or component also presents distinct junctures for assessment, audit, and other measures. How should AI accountability mechanisms consider the AI lifecycle?
    b. How should AI audits or assessments be timed? At what stage of design, development, and deployment should they take place to provide meaningful accountability?

An advanced AI that might possible overpower humans at the first opportunity cannot be safely tested. If, during the test, it is possible for it to thwart human control and gain extensive power, then the test cannot be safe. If, on the other hand, such overpowering is not possible during the test, because the AI's interaction with its test environment is too limited, then the test is not valid. Even "internal" use of such an AI (as opposed to "deployment") is not safe in the first place. Therefore, for the risk of extinction, audits and assessments must mainly occur after the design stage, but before the development phase. (Before the design stage, there is little to assess.) It is possible that automated assessments could also be incorporated in early stages of development, with any failure resulting in the immediate termination of the project.

17. How should AI accountability measures be scoped (whether voluntary or mandatory) depending on the risk of the technology and/or of the deployment context? If so, how should risk be calculated and by whom?

For the riskiest technology, accountability measures should be mandatory. There should be little disagreement about how bad extinction would be. We suspect that most legislators' constituents would not accept any AI project for which this outcome is plausible.

20. What sorts of records (e.g., logs, versions, model selection, data selection) and other documentation should developers and deployers of AI systems keep in order to support AI accountability? How long should this documentation be retained? Are there design principles (including technical design) for AI systems that would foster accountability-by-design?

For organizations and individuals to be held accountable, they must auditable. That means they must keep track of what code they run and with what arguments. They must keep a record of the data and calculate the total computation used to train any model. Often, the function of code is difficult to understand simply by reading it, especially without broader context and mathematical background. It may only be feasible to ascertain the purpose and function of different programs by referring to associated internal communication, so organizations could be required to keep records of such communication, alongside documentation of code. The value of this information makes it well worth retaining for decades, or longer if certain models are still in use. It may be wise for the government to store code on companies' behalf, especially if any companies go bankrupt.

Building a long-term planning agent can happen in pieces. If one group extensively trains a powerful predictive model, they could allow another group or individual to build an agent that plans its actions using that predictive model. Therefore, all powerful predictive models should be registered and annotated with the amount of compute used to train them (alongside any other legally relevant information), so that others can understand the legality of incorporating them into a long-term planning agent. A long-term planning agent can also be built by composing multiple short-term planning agents (if the short-term planning agents have flexible goals rather than fixed ones)—the way this is works is you have a short-term agent accomplish the short-term goal of putting the next short-term agent in a position that it considers favorable for its short-term goal. So short-term agents with flexible goals should be registered and annotated with a report of how far ahead they can plan (alongside any other legally relevant information); if any are composed, these numbers must be added.

The law must be robust to the possibility of federated learning, the practice of distributing machine learning algorithms over many different computers. This can occur at a much lower level than the examples above. To address this, any time any information relevant to a machine learning algorithm is sent from one computer to another (like gradient updates to parameters), it must be accompanied by legally relevant information about the process used to produce that information. The same applies to “synthetic data” made by generative AI for the purpose of training another predictive model.

21. What are the obstacles to the flow of information necessary for AI accountability either within an organization or to outside examiners? What policies might ease researcher and other third-party access to inputs necessary to conduct AI audits or assessments?

Code is difficult to read, even if you already know what it does. Internal communications and documentation may be required for auditors to understand the function of code. The documentation should be detailed enough to allow an auditor to verify that the code does what the documentation claims.

24. What are the most significant barriers to effective AI accountability in the private sector, including barriers to independent AI audits, whether cooperative or adversarial? What are the best strategies and interventions to overcome these barriers?

Cost and potential lack of regulatory expertise are major barriers to wide-scale accountability. One potential remedy is to offer bounties for whistleblowers, which could encourage technical experts within an organization to investigate and report legal violations.

Another barrier is the proprietary nature of organizations' machine learning algorithms (as noted by question 27); their internal records could not simply be made public for any bounty hunter to investigate. Perhaps the law could allow certain aspiring bounty hunters with security clearance to sign non-disclosure and non-compete agreements and thereby be granted special access to such records in search of violations. If AI is widely developed by individuals, a bounty system may be the only way to cope with the scale of the activity that needs auditing.

26. Is the lack of a federal law focused on AI systems a barrier to effective AI accountability?

Absolutely.

30. What role should government policy have, if any, in the AI accountability ecosystem? For example:

    a. Should AI accountability policies and/or regulation be sectoral or horizontal, or some combination of the two?

For AI policy as a whole, likely a combination of the two. But for the purposes of avoiding human extinction, horizontal.

    b. Should AI accountability regulation, if any, focus on inputs to audits or assessments (e.g., documentation, data management, testing and validation), on increasing access to AI systems for auditors and researchers, on mandating accountability measures, and/or on some other aspect of the accountability ecosystem?

All of the above. We may need the whole regulatory toolkit to address the risk of human extinction.

c. If a federal law focused on AI systems is desirable, what provisions would be particularly important to include? Which agency or agencies should be responsible for enforcing such a law, and what resources would they need to be successful?

It would be important to include provisions which prevent the development of advanced artificial agents planning over the long term, enforced with a rigor that is proportional to the significant risk of human extinction. Ideally, a new agency would be created that is solely responsible for preventing AI-based extinction. But a broader agency to oversee all regulation of AI may be acceptable. They would need to employ highly paid technical experts and economists, and they may need the ability to outsource enforcement tasks through the use of bounties.

d. What accountability practices should government (at any level) itself mandate for the AI systems the government uses?

For accountability practices designed to prevent human extinction, these must apply to the government as well. The only difference is that certain usage of AI by the government in highly sensitive areas may require auditors to have Top Secret clearance.

33. How can government work with the private sector to incentivize the best documentation practices?

Legal mandates may be necessary. Small fines might be levied if documentation is vague enough that multiple third parties interpret it differently.

34. Is it important that there be uniformity of AI accountability requirements and/or practices across the United States? Across global jurisdictions? If so, is it important only within a sector or across sectors? What is the best way to achieve it? Alternatively, is harmonization or interoperability sufficient and what is the best way to achieve that?

Uniformity across global jurisdictions, and across sectors, may be necessary. International coordination, if not perfect uniformity, would likely be necessary at the very least. Existentially dangerous AI could originate anywhere. The best way to achieve this may be by speaking softly and carrying a big stick.

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